

Weightfield2

User Manual

Build 8.0 — “The Claude Edition”

Simulation of Semiconductor Particle Detectors
Silicon Strip Detectors and Ultra-Fast Silicon Detectors (UFSD)

February 2026

Based on source code analysis of Weightfield2 Build 8.0

0 Contents

1 Introduction

Weightfield2 is a simulation program for semiconductor particle detectors, developed for research in high-energy physics instrumentation. It is particularly suited for the study of **silicon strip detectors** and **Ultra-Fast Silicon Detectors (UFSD)**, which are thin silicon sensors equipped with an internal gain layer that provide extremely fast timing signals (down to a few tens of picoseconds).

The program simulates the full chain of physical phenomena that occur when a charged particle (or photon) traverses a semiconductor detector:

- The electrostatic potential distribution inside the detector (drift and weighting potentials), computed by solving the Poisson equation on a 2D grid.
- The drift and diffusion of charge carriers (electrons and holes) through the resulting electric field.
- The current induced on the readout electrode, calculated via the Shockley–Ramo theorem.
- Optional effects: impact ionization (gain), charge trapping due to irradiation, magnetic field (Lorentz angle), space-charge repulsion, and thermal diffusion.
- The response of a configurable readout electronics chain (Charge-Sensitive Amplifier or Transimpedance Amplifier), including noise, threshold crossing time, jitter, and time-over-threshold.

The program is built on the **ROOT framework** (CERN) for the graphical user interface, histogramming, and visualization. It is written in C++ and runs on macOS and Linux.

The name “Weightfield” derives from the central role of the *weighting field* in computing the induced current via the Shockley–Ramo theorem.

2 Physics Background

2.1 Semiconductor Detectors

A silicon strip detector is a reverse-biased p-n junction. When a charged particle traverses the silicon, it loses energy by ionization, creating electron-hole (e-h) pairs along its track (approximately 75 e-h pairs per micron of silicon at minimum ionizing particle (MIP) energy). The applied bias field drifts electrons toward one electrode and holes toward the other. The motion of these charges induces a measurable current pulse on the readout electrode.

2.2 Electrostatic Potential

The electrostatic (drift) potential $\varphi_{\text{drift}}(x, y)$ inside the detector satisfies the **Poisson equation**:

$$\nabla^2 \varphi_{\text{drift}} = -\frac{\rho}{\varepsilon} \quad (1)$$

where ρ is the space charge density (from ionized dopants) and $\varepsilon = \varepsilon_0 \varepsilon_r$ is the permittivity of the material ($\varepsilon_r = 11.9$ for silicon). The electric field is:

$$\mathbf{E} = -\nabla\varphi_{\text{drift}} \quad (2)$$

2.3 Weighting Potential

To compute the induced current, the **Shockley–Ramo theorem** requires a *weighting potential* $\varphi_w(x, y)$, which is the solution to the Laplace equation (no space charge) with the readout electrode set to unity and all other electrodes grounded:

$$\nabla^2\varphi_w = 0, \quad \varphi_w|_{\text{readout}} = 1, \quad \varphi_w|_{\text{others}} = 0 \quad (3)$$

2.4 Induced Current: Shockley–Ramo Theorem

The instantaneous current induced on the readout electrode by a carrier of charge q moving with velocity $\mathbf{v}$ is:

$$i(t) = q \mathbf{v}(t) \cdot \mathbf{E}_w(x(t), y(t)) \quad (4)$$

where $\mathbf{E}_w = -\nabla\varphi_w$ is the weighting field. The total induced charge from a carrier drifting from its creation point to an electrode is:

$$\Delta Q = q [\varphi_w(\text{final}) - \varphi_w(\text{initial})] \quad (5)$$

3 Installation and Running

3.1 Prerequisites

- **ROOT** framework (version 6 recommended), compiled with GUI support.
- A C++11 (or later) compiler (`clang++` on macOS, `g++` on Linux).

3.2 Compilation

```
make
```

This produces the executable `weightfield`.

3.3 Running

```
./weightfield
```

The program looks for a parameter file named `parameters.dat` in the `sensors/data/` directory. If not found, built-in default values are used and a warning is printed.

To specify a custom parameter file:

```
./weightfield mysimulation.dat
```

If a `.dat` file is given as argument, the program also *loads* the previously saved simulation state from that file automatically at startup.

3.4 Alternative: ROOT macro

```
root run_wf.C
```

3.5 Post-processing

```
root WF_ReadBack.C
```

Reads and plots current waveforms, gain distributions, and timing from previously saved output files.

4 The Parameter File

Weightfield2 reads initial parameter values from a plain-text file (default: `sensors/data/parameters.dat`). Each non-empty line contains a keyword followed by its numerical value. Lines starting with `#` are comments and are ignored.

Keyword	Default	Description
STRIP_NUMB	3	Number of strips simulated (must be odd: 1, 3, 5...)
DETECT_HEIGHT	200	Detector thickness [μm]
STR_PITCH	300	Strip pitch (center-to-center) [μm]
STR_WIDTH	200	Strip implant width [μm]
GAIN_SCL	1.3	Gain scale factor (used with Force Gain)
SGAIN_SCL	1	Screened gain scale factor
DOP_LEV	1.2	Gain implant peak doping [$10^{16}/\text{cm}^3$]
GAIN_LYR_RSS	10	Gain layer recess from strip edge [μm]
YPOSITION	100	Y position for edge/X-ray injection [μm]
IRRADIATION	10	Neutron fluence [$10^{14} \text{ n}_{\text{eq}}/\text{cm}^2$]
BETA_ELECTRONS	4.9	Trapping β for electrons
BETA_HOLES	6.2	Trapping β for holes
DOUBLEJUNCTION	1	Double-junction N_A parameter
NA_OVER_ND	8	Double-junction C_D parameter
CA	4	Double-junction C_A parameter
QUENCH	1	Gain quenching coefficient
PRINT	40	Output file sampling rate [GS/s]
BIAS_VOLTAGE	200	Bias voltage [V]
DEPL_VOLTAGE	10	Bulk depletion voltage [V]
CAPACITANCE	2	Detector capacitance [pF]
INDUCTANCE	2	RC time constant [ns]
OSCOPE_BW	2	Oscilloscope bandwidth [GHz]
IMPEDANCE	50	CSA input impedance [Ω]
SHPR_INT_TIME	3.5	CSA rise time (10–90%) [ns]
SHPR_DCY_TIME	5	CSA fall time [ns]

continued on next page

Keyword	Default	Description
SHPR_TRANS	4	CSA transimpedance [mV/fC]
SHPR_NOISE	1	CSA noise [mV]
VTH	10	CSA threshold [mV] (if < 1: CFD fraction)
BBBW	2.5	TI amplifier bandwidth [GHz]
BBGAIN	100	TI gain [k Ω]
BBVTH	10	TI threshold [mV]
BB_NOISE	2	TI noise [mV]
BB_IMP	50	TI input impedance [Ω]
PRECISION	5	Carrier thinning factor (1=full, 100=fastest)
SAMPLING	1	Time step [ps]
STEPX	1	Mesh step along x [μm]
STEPLY	0.1	Mesh step along y [μm]
USERQ	70	User-defined charge [e-h/ μm]
NUMBERP	1	Number of particles per simulation
TEMPERATURE	300	Temperature [K]
DISPERSION	1	Space-charge cloud scale factor
DOPING_GL	0	Gain layer doping type (combo index)
KIND_GL	0	Gain mechanism index
SHAPE_GL	0	Gain layer shape index

5 The Graphical Interface — Overview

The Weightfield2 window is divided into two main areas arranged horizontally:

- **Left:** a tabbed display panel showing all computed quantities as plots and graphs.
- **Right:** the control panel organized in two vertical columns with all simulation parameters and action buttons.

A green status bar labeled “*Press 'Potential' and then 'Currents'*” sits above the control buttons. It turns **red** during computation and returns to **green** when idle. The window title reads: *Weightfield2 Build 8.0 The Claude Edition*.

6 Display Tabs (Left Panel)

The left panel contains five tabs selectable from the top tab bar.

6.1 Drift Potential Tab

Displays the electrostatic (drift) potential computed by the Poisson solver.

Top canvas A 2D colour map (TH2F) of $\varphi_{\text{drift}}(x, y)$ across the full detector cross-section. The x -axis spans the total simulated width (number of strips $\times$ pitch); the y -axis spans the detector thickness from the readout surface (top) to the backplane (bottom). The colour encodes the potential in Volts.

Bottom canvas Divided into a 2×2 grid of sub-pads:

1. 1D cut of the drift potential along a vertical line at the selected x position.
2. 1D cut of the drift field $|E_y(y)|$ at the same position.
3. Doping density profile vs. depth.
4. Impact ionization gain as a function of depth (visible only when a gain layer is active).

Controls below the canvas

Control	Function
On Strips (red)	Sets the 1D cut position to the centre of a readout strip.
Between Strips (blue)	Sets the 1D cut position to the midpoint between two strips.
Number entry	Manual entry of the x -coordinate [μm] for the 1D cut.
Draw	Draws the 1D cut at the manually entered x position.
Ex	Replaces the field display with the horizontal component $E_x(x, y)$.
Ey	Replaces the field display with the vertical component $E_y(x, y)$.

All these buttons are *enabled only after potentials have been calculated*.

6.2 Weighting Potential Tab

Displays the weighting potential $\varphi_w(x, y)$ (dimensionless, ranging from 0 to 1). The layout mirrors the Drift Potential tab.

Top canvas 2D colour map of $\varphi_w(x, y)$.

Bottom canvas 1D cuts of the weighting potential and weighting field.

The field buttons are labelled $|E_x|$ and $|E_y|$ (absolute values of the weighting field components).

6.3 Currents and Oscilloscope Tab

The main output tab. Displays all current waveforms and integrated charge.

Current canvas

Shows $I(t)$ in Amperes vs. time in nanoseconds. The contributions are colour-coded:

Colour	Contribution
Red	Primary electron current
Magenta	Gain electron current (from impact ionization)
Blue	Primary hole current
Cyan	Gain hole current
Green	Total current (sum of all contributions)
Dashed	Current at CSA input (after RC filtering, if enabled)
Thin line	Oscilloscope output (after full electronics chain)

Charge Collection panel

Displays the integrated charge after the simulation in units of 10^3 elementary charges (ke):

Label	Meaning
e^- charges (e)	Charge collected from primary electrons
h^+ charges (e)	Charge collected from primary holes
$e^- + h^+$ charges (e)	Total primary charge
Gain e^- charges (e)	Electrons from impact ionization
Gain h^+ charges (e)	Holes from impact ionization
Gain $e^- + h^+$ charges (e)	Total gain charge
Total e^- charges (e)	Primary + gain electrons
Total h^+ charges (e)	Primary + gain holes
Total Charges (e)	Grand total collected charge

Lorentz Drift panel

Shows the Lorentz angle in degrees for electrons and holes (non-zero only when the magnetic field is enabled).

Carriers lifetime panel

Shows the trapping lifetime τ_e and τ_h in nanoseconds for the current fluence setting.

6.4 Electronics I Tab

Displays the amplified signal in mV vs. time, after passing through the configured electronics chain. In batch mode, accumulates:

- Jitter histogram (time resolution) for both CSA and TI channels.
- Time-over-threshold (ToT) scatter plot vs. arrival time.
- Time resolution histogram.

6.5 Electronics II Tab

Displays additional histograms in batch mode, including the Landau distribution of deposited charge and the distribution of measured e-h pairs.

7 Control Panel — Left Column

7.1 Run Frame

Located at the top of the control panel. Contains five action buttons:

Button	Function
Set	Reads all detector geometry parameters from the GUI, sets the Poisson grid boundary conditions, and initialises the doping profile. Must be pressed before Potentials .
Potentials	Starts the potential calculation in a background thread. Solves the Poisson equation for the drift potential and the Laplace equation for the weighting potential. Also computes the electric field.
Currents	Starts the current calculation. Generates charge carriers, drifts them through the fields, and computes induced currents. Enabled only after potentials have been calculated.
Stop (red)	Interrupts the currently running calculation gracefully.
Exit (green)	Closes the window and terminates the program.

Typical run sequence: Set → Potentials → Currents

7.2 Precision Frame

Control	Default	Function
eh pairs followed (1=Most precise, 100=Fastest)	5	Carrier thinning factor. Only 1-in-N carrier pairs is tracked individually; the rest contribute scaled-up charge. Value 1 = full precision (slow). Value 100 = fastest (noisy).
Time Step [ps]	1	Integration time step for carrier drift. Smaller = more accurate, slower.
Mesh x [μm]	1	Spatial grid step in the strip direction.
Mesh y [μm]	0.1	Spatial grid step in the depth direction. Automatically refined if the gain layer is not well sampled.

7.3 Output Files Frame

Control	Default	Function
ON	Off	Enables writing signal waveforms to a text file. When enabled, the filename and sampling rate become active.
GS/s	40	Sampling rate for the output file in Gigasamples per second.
Name	wf	Base name for output text files. Files are written as wf_0.txt, wf_1.txt, ... in sensors/data/.

7.4 Batch Mode Frame

Allows automatic repetition of the current calculation for many events to accumulate statistics.

Control	Default	Function
ON	Off	Enables batch mode. The Currents calculation repeats automatically for the specified number of events.
Starting at event	0	First event number (for file naming and random seed offset).
for events	1	Total number of events to simulate.
Rnd	Off	When checked, each event uses a random x -position for the particle impact (uniform across one strip pitch).

7.5 Select Particles Frame

Defines the type of ionizing particle or radiation simulated.

Particle Type Dropdown

Option	Description
MIP: uniform Q	MIP with constant charge density along the full detector thickness. Pairs-per-micron set in the specifics entry.
MIP: NON uniform Q	MIP with non-uniform (Bethe–Bloch-like) energy deposition profile.
MIP Landau	MIP with Landau-distributed total charge per event. Most realistic model for single-event simulation.
Laser (1064 nm): Top-TCT	IR laser from the readout side. Absorption length ~ 1 mm. Simulates TCT measurements.
Laser (1064 nm): Edge-TCT	IR laser entering horizontally from the side at depth Y [μm]. Simulates Edge-TCT.
Edge MIP Landau	MIP entering horizontally from the side.
Laser (4–600 nm)/Alpha from top	Short-wavelength laser or 5 MeV alpha particle from the readout side. Range ~ 30 μm in silicon.
Laser (4–600 nm)/Alpha from bottom	Same, from the backplane side.
X-Ray	X-ray absorbed at a single point. Depth drawn from exponential distribution with mean set by photon energy.
Current pulse	Fixed charge injected at specified (x, y) with specified duration.

Specifics Entry

The numerical entry below the dropdown changes meaning with the particle type:

Particle type	Entry meaning
MIP (all)	#e-h/ μm — pairs per micron (most probable value for Landau)
Laser top/edge	#e-h/ μm — pairs per micron of laser path
Alpha / short laser	Range [μm] — penetration depth
X-Ray	E [keV] — photon energy (determines absorption depth)
Current pulse	Duration [ns] or charge [ke]

Position Controls

Control	Function
X [μm], Angle [D]	x -position of the particle impact point and track angle in degrees (0° = vertical).
Y [μm]	Depth for edge-TCT or X-ray injection from the top surface.
Number of Particles	Number of independent particles per Currents calculation (charges superimposed).

7.6 Irradiation Frame

Implements radiation damage effects on the detector.

Top Row

Control	Default	Function
All On/Off	Off	Master switch enabling/disabling all irradiation sub-options.
Fluence [$10^{14} \text{ n}_{\text{eq}}/\text{cm}^2$]	10	Neutron-equivalent fluence for trapping and acceptor creation.
Irradiation particle	neutrons	Selects neutrons or protons/pions, affecting the β trapping coefficients.

Effect Checkboxes

Checkbox	Function
Gain quenching	Reduces gain layer doping proportionally to fluence, simulating radiation-induced boron removal.
Charge trapping	Enables finite carrier lifetime $\tau = 1/(\beta\Phi)$. Reduces collected charge by trapping.
Quench coefficient	Scale factor for the gain quenching rate per unit fluence.
$\ln(\Phi > 5 \times 10^{15})$	Switches the acceptor creation model to logarithmic dependence at very high fluences.
Acceptor creation	Models radiation-induced acceptor creation in the bulk (Hamburg model), shifting the effective doping.
Doping rem.	Simulates initial donor removal (beneficial annealing phase).

Compensation (Double Junction)

Control	Function
Compensation	Enables a space-charge compensation model where radiation-induced donors and acceptors partially cancel.
N_A	Acceptor density parameter in the compensation model.
C_D	Donor compensation coefficient.
C_A	Acceptor creation coefficient.

7.7 Drawing Frame

Checkbox	Function
Equipotential Lines	Overlays equipotential contours on the 2D potential colour maps.
Current Absolute Value	Displays $ I(t) $ instead of $I(t)$.
e-h Motion	Shows carrier positions during drift (animation/snapshot).

7.8 Current Settings Frame

Control	Default	Function
B-Field on	Off	Enables the magnetic field calculation and the Lorentz angle.
Tesla (positive = into page)	0	Magnetic field strength [T]. Positive = field into the (x, y) plane.
Diffusion	Off	Enables thermal diffusion. Carriers receive a random transverse kick at each step with $\sigma = \sqrt{2D\Delta t}$.
Space charge effects	Off	Simulates Coulomb repulsion within the charge cloud. Not applicable for alpha particles.
Dispersion factor	1	Scale factor for the space-charge cloud size.
Temperature [K]	300	Operating temperature. Affects mobilities, saturation velocities, and diffusion coefficient.

8 Control Panel — Right Column

8.1 Files: Save and Load

Control	Function
Save (green)	Saves the complete simulation state (all parameters and computed potentials) to the named .dat file in sensors/data/ . Allows resuming without recomputing potentials.
Load (green)	Restores a previously saved simulation state. After loading, Currents can be run directly.
File name entry	Name of the file to save to or load from.
SaveGraph	Saves the currently displayed graphs to ROOT files in sensors/graph/ .
Special WF file?	When checked, saves an additional file compatible with WeightField analysis scripts.

8.2 Detector Properties Frame

Type

Selects the semiconductor material:

Button	μ_e [m ² /Vs]	μ_h [m ² /Vs]	E_{ion} [eV]	Notes
Si	0.1417	0.0470	3.6	Standard detector material
Diamond	0.18	0.15	~13	Radiation hard, large band gap
SiC	0.08	0.0115	8.0	Wide band gap

Doping

Two independent radio-button groups:

- **Strips:** n or p — doping type of the readout strip implants.
- **Bulk:** n or p — doping type of the sensor bulk.

Standard configurations: **n-in-p** (UFSD default), **p-in-n** (classic strip detector).

Dimensions

Control	Default	Function
# of strips (1, 3, 5...)	3	Number of strips simulated. Must be odd.
Thickness [μm]	200	Detector thickness.
Implant width [μm]	200	Width of the readout strip implant. Must be < pitch.
Strip pitch [μm]	300	Center-to-center strip distance.

If the implant width exceeds the pitch, the program automatically sets: $w = \text{pitch} - 10 \mu\text{m}$.

Voltage

Control	Default	Function
Bias [V]	200	Total reverse bias voltage.
Bulk Depletion [V]	10	Full depletion voltage of the bulk, used to compute the bulk doping density.

Gain Frame

Configures the internal gain layer (LGAD-type detectors).

	Option	Description
Gain Doping Dropdown	No gain layer	Standard detector, no multiplication.
	Boron	Standard p-type gain implant.
	Boron + Carbon	Carbon co-implantation reduces boron diffusion enabling thinner layers.
	Comp32	Specific boron-carbon recipe (RD50 collaboration).
	Comp32 + Carbon	Comp32 with additional carbon.

Gain Shape Dropdown Selects the spatial position of the gain layer within the detector (depth from readout surface):

Label	Depth range	Notes
I	0.2 – 0.7 μm	Shallow implant
II	0.5 – 1.0 μm	
III	1.0 – 1.4 μm	
IV	1.1 – 1.5 μm	
V	1.3 – 1.9 μm	
VI	1.6 – 2.1 μm	
VII	1.8 – 2.2 μm	Wide / deep
VIII	0.0 – 3.0 μm	
IX	3.0 – 3.5 μm	Deep implant
X	Near ohmic contact	Inverted LGAD geometry

Gain Mechanism Dropdown Selects the impact ionization model:

Option	Description
Gain Mechanism = OFF	No multiplication.
Massey – WF2 LGAD model	Default LGAD model in Weightfield2.
Massey optimized	Re-fitted Massey coefficients.
van Overstraeten optimized	Optimised van Overstraeten – de Man model.
Okuto – Crowell model	Classic parametrization.
Bologna model	Temperature-dependent model from the Bologna group.
van Overstraeten – de Man	Original 1970 model.

	Control	Default	Function
Gain Parameters	Peak doping [$10^{16}/\text{cm}^3$]	1.3	Peak gain layer doping. Higher = stronger local field = higher gain.
	Gain scale factor	1.3	Multiplicative gain override (active with Force Gain).
	High precision	Off	Refines the grid until the gain layer is sampled to within 1.5%.
	Force Gain	Off	Applies the manual gain scale factor, overriding the automatic computation.
	Screened Gain	Off	Electric field screening by the mobile charge cloud.
	Recess [μm]	10	Lateral gap between strip edge and gain layer (prevents edge breakdown).

8.3 Electronics Frame

Readout

Option	Description
Top – DC	Readout from the strip side, DC-coupled. Full current pulse to amplifier.
Top – AC	Readout from the strip side, AC-coupled through a coupling capacitor. Signal is differentiated.
Back	Readout from the backplane.

ON Checkbox and CSA Kind Dropdown

The **ON** checkbox activates the electronics simulation. The **CSA Kind** dropdown selects presets:

Option	Description
Generic CSA	Fully user-configurable CSA.
CSA: ETROC	Preset for the CMS ETL ETROC ASIC.
CSA: NA62	Preset for the NA62 frontend.
CSA: FAST3	FAST3 ASIC preset.
TI: FAST low gain	FAST ASIC, transimpedance mode, low gain.
TI: FAST mid gain	FAST ASIC, medium gain.
TI: FNAL Board	FNAL transimpedance amplifier board.
TI: SC	Single-channel TI preset.
TI: FAST EVO high gain	FAST EVO ASIC, high gain mode.

CSA Parameters

Parameter	Default	Meaning
Cdet [pF], RC [ns]	2, 2	Detector capacitance and bias network RC time constant.
Scope BW [GHz]	2	Oscilloscope ($50\ \Omega$) bandwidth.
CSA Impedance [Ω]	50	CSA input impedance.
CSA Transimpedance [mV/fC]	4	Voltage gain per unit charge.
CSA rise time [ns]	3.5	10–90% rise time at $C_{\text{det}} = 0$.
CSA fall time [ns]	5	Shaper decay time.
CSA noise [mV]	1	RMS noise added to the CSA output.
CSA threshold [mV]	10	Discriminator threshold. If < 1 : CFD fraction.

TI (Transimpedance) Parameters

Parameter	Default	Meaning
TI impedance [Ω]	50	Input impedance.
TI bandwidth [GHz]	2.5	3 dB bandwidth of the amplifier.
TI gain [k Ω]	100	Transimpedance gain.
TI noise [mV]	2	RMS noise added to the TI output.
TI threshold [mV]	10	Discriminator threshold. If < 1 : CFD fraction.

9 Physics Models in Detail

9.1 Poisson Equation and Multigrid Solver

The 2D drift potential is solved on a rectangular grid indexed by (i, j) where i runs along x (strip direction) and j along y (depth). Grid spacings are BinSize $_x$ and BinSize $_y$ in $\mu\text{m}/\text{bin}$.

Boundary conditions:

- Readout electrode: $\varphi = V_{\text{bias}}$ (or 0, depending on configuration).
- Opposite electrode (backplane or strips): $\varphi = 0$ (or V_{bias}).
- Left/right boundaries: periodic (the simulation domain repeats laterally with the period of the strip pitch).

The **multigrid algorithm** (V-cycle) is used for fast convergence:

1. Apply a few Gauss–Seidel relaxation sweeps at the finest grid.
2. *Restriction*: compute the residual and transfer it to a coarser grid (XMAX/2, YMAX/2).
3. Repeat recursively on progressively coarser grids.

4. *Prolongation*: interpolate the correction back to the finer grid.
5. Smooth again at the finer level.

The number of multigrid levels is determined automatically:

$$N_{\text{multig}} = \left\lfloor \frac{\log(\text{XMAX} \cdot \text{YMAX}/3)}{\log 2} \right\rfloor + 1 \quad (6)$$

Grid dimensions are adjusted (padded to be odd and compatible with repeated halving) to ensure each coarsening step produces a grid with an odd number of points in each dimension.

The **doping profile** is initialized on the same grid. For standard detectors: uniform bulk doping N_D computed from V_{depl} . For LGAD: a Gaussian or step-function gain implant is added near the readout surface.

9.2 Electric Field

The field components are computed by centred finite differences:

$$E_x[j][i] = -\frac{\varphi[j][i+1] - \varphi[j][i-1]}{2 \Delta x \times 10^{-6}} \quad [\text{V/m}] \quad (7)$$

$$E_y[j][i] = -\frac{\varphi[j+1][i] - \varphi[j-1][i]}{2 \Delta y \times 10^{-6}} \quad [\text{V/m}] \quad (8)$$

9.3 Carrier Transport: Drift and Diffusion

Each carrier is stepped through the field with timestep Δt [ps]. At each step:

Drift velocity (velocity saturation model):

$$v_{\text{drift}} = \frac{\mu |\mathbf{E}|}{\sqrt{1 + \left(\frac{\mu |\mathbf{E}|}{v_{\text{sat}}} \right)^2}} \quad (9)$$

This interpolates between the Ohmic regime ($v \propto E$) and saturation ($v \rightarrow v_{\text{sat}}$).

Diffusion (if enabled): A Gaussian random displacement is added:

$$\sigma = \sqrt{2 D \Delta t}, \quad D = \frac{\mu k_B T}{e} \quad (10)$$

Trapping (if irradiation enabled): At each step, the carrier has trapping probability:

$$P_{\text{trap}} = 1 - e^{-\Delta t/\tau}, \quad \tau = \frac{1}{\beta \Phi} \quad (11)$$

Trapped carriers cease to move and contribute no further to the current.

9.4 Carrier Mobility and Velocity Saturation

Temperature-dependent mobility (Sentaurus model):

$$\mu_e(T) = \mu_{e,300} \left(\frac{T}{300} \right)^{-2.5} \quad (12)$$

$$\mu_h(T) = \mu_{h,300} \left(\frac{T}{300} \right)^{-2.2} \quad (13)$$

At 300 K for silicon: $\mu_{e,300} = 0.1417 \text{ m}^2/(\text{V}\cdot\text{s})$, $\mu_{h,300} = 0.0470 \text{ m}^2/(\text{V}\cdot\text{s})$.

Saturation velocity (Canali 1975):

$$v_{\text{sat},e}(T) = 1.07 \times 10^5 \left(\frac{300}{T} \right)^{0.87} \text{ [m/s]} \quad (14)$$

$$v_{\text{sat},h}(T) = 8.37 \times 10^4 \left(\frac{300}{T} \right)^{0.52} \text{ [m/s]} \quad (15)$$

Fluence-dependent mobility reduction (Mandi 2020):

$$(\mu_e + \mu_h)_{\text{irr}} = \min \left(\mu_e + \mu_h, \frac{0.35}{(\Phi/10^{15})^{0.46}} \right) \text{ [m}^2/(\text{V} \cdot \text{s})] \quad (16)$$

9.5 Lorentz Angle

In a magnetic field B perpendicular to the detector plane, the Lorentz angle is:

$$\tan \theta_L = \mu_H \cdot B \quad (17)$$

where $\mu_H \approx r_H \cdot \mu$ is the Hall mobility ($r_H \approx 1.15$ for electrons, ≈ 1 for holes in silicon). Carriers drift at angle θ_L to the electric field, causing lateral displacement of the charge cloud and inter-strip charge sharing.

9.6 Signal Induction: Shockley–Ramo Theorem

The current induced by a carrier of charge q at position $(x(t), y(t))$ with velocity $\mathbf{v}$ is (Eq. ??). The current is integrated numerically:

$$I_{\text{total}}(t) = \sum_{\text{carriers}} q_k \mathbf{v}_k(t) \cdot \mathbf{E}_w(x_k(t), y_k(t)) \quad (18)$$

The total collected charge from all carriers:

$$Q_{\text{tot}} = \int_0^\infty I_{\text{total}}(t) dt = \sum_k q_k [\varphi_w(\text{electrode}) - \varphi_w(\text{creation point})] \quad (19)$$

9.7 Impact Ionization and Gain

When the electric field in the gain layer exceeds $\sim 10^5$ V/cm, carriers gain sufficient energy for impact ionization. The ionization rate is described by:

$$\alpha = A \exp\left(-\frac{B}{E}\right) \quad [\text{cm}^{-1}] \quad (20)$$

Different models provide different parametrizations $A(T)$, $B(T)$ for electrons and holes.

In Weightfield2: A carrier accumulates “leftover gain potential” as it traverses the gain layer. When this exceeds the threshold for one ionization event, a secondary e-h pair is created and tracked. Secondary carriers also generate gain (avalanche). Gain carriers are flagged separately and shown in the magenta/cyan waveforms.

Gain quenching due to irradiation:

$$N_{\text{eff,gain}} = N_{\text{gain},0} \exp(-c \Phi) \quad (21)$$

where c is the quenching coefficient, tunable via the Quench parameter.

9.8 Irradiation Effects

Carrier trapping:

$$\frac{1}{\tau_e} = \beta_e \Phi, \quad \frac{1}{\tau_h} = \beta_h \Phi \quad (22)$$

For neutrons: $\beta_e \approx 4.9 \times 10^{-16} \text{ cm}^2/\text{ns}$, $\beta_h \approx 6.2 \times 10^{-16} \text{ cm}^2/\text{ns}$.

Acceptor creation (Hamburg model): Radiation creates acceptors in the bulk, shifting the effective space charge and changing V_{depl} . For very high fluences ($\Phi > 5 \times 10^{15}$), the logarithmic model modifies the rate of increase.

Double junction: At high fluence, both donors and radiation-induced acceptors coexist, leading to a complex space-charge profile (double junction). The Compensation section of the GUI models this phenomenon.

9.9 Supported Materials

Material	ε_r	E_{ion} [eV]	μ_e [m^2/Vs]	$v_{\text{sat},e}$ [m/s]	$v_{\text{sat},h}$ [m/s]
Silicon (Si)	11.9	3.6	0.1417	1.07×10^5	8.37×10^4
Diamond (C)	~ 5.7	~ 13	0.18	2.7×10^5	2.7×10^5
SiC	~ 9.7	8.0	0.08	2.7×10^5	2.7×10^5

10 Electronics Chain

After computing the raw current $I(t)$ from the Shockley–Ramo theorem, the electronics simulation proceeds as follows:

1. **RC filtering:** The current passes through the detector–bias network, modelled by detector capacitance C_{det} and RC time constant. This shapes the signal at the amplifier input.
2. **Oscilloscope bandwidth:** A Gaussian low-pass filter with cut-off at the specified bandwidth limits the signal. Models the ADC or oscilloscope.
3. **CSA:** Integrates the input current. Output voltage:

$$V_{\text{CSA}}(t) = Q(t) \times \text{transimpedance} \quad (23)$$

Rise time T_r (or $C_{\text{det}} \times Z$, whichever is larger), fall time T_f , Gaussian noise σ_{noise} added.

4. **TI (Broadband):** Amplifies the current. Bandwidth-limited Gaussian response. Gain = BBGain [k Ω].
5. **Threshold / CFD:** Time of threshold crossing recorded. If $V_{\text{th}} > 1 \text{ mV}$: fixed discriminator. If $V_{\text{th}} < 1$: constant fraction discriminator (CFD) at fraction $V_{\text{th}} \times V_{\text{peak}}$.

Time resolution estimate:

$$\sigma_t \approx \frac{\sigma_{\text{noise}}}{(dV/dt)|_{\text{threshold}}} \quad (24)$$

In batch mode, the distribution of threshold-crossing times is histogrammed (Electronics I tab) to give the full time resolution including Landau fluctuations.

11 Output Files and Plots

In-GUI plots Any canvas can be saved via the ROOT right-click context menu in PNG, PDF, ROOT format.

Saved state files (sensors/data/) Pressing **Save** writes all parameters and computed potentials to a .dat file. **Load** restores the full state.

Signal waveforms (sensors/data/) When “Output files for signals” is ON: two-column text files (time [ns], current [A]) at the specified sampling rate.

Graph files (sensors/graph/) ROOT files with all graph objects, saved by **SaveGraph**.

WF_ReadBack.C ROOT macro for offline analysis: reads signal files, computes mean waveforms, collected charge, gain estimates, and timing distributions.

12 Typical Workflow

12.1 Basic Silicon Strip Detector

1. Set geometry: 3 strips, 200 μm thick, 300 μm pitch, 150 μm implant, n-in-p.
2. Set Bias = 200 V, Depletion = 50 V.

3. Particle: MIP Landau, ~ 75 e-h/ μm MPV.
4. **Set** $\rightarrow$ **Potentials** $\rightarrow$ inspect the 2D maps and 1D cuts.
5. **Currents** $\rightarrow$ read collected charge from the Charge Collection panel.

12.2 LGAD Simulation

1. Detector: 50–200 μm , Si, n-in-p.
2. Gain doping: Boron. Gain shape: III (1.0–1.4 μm). Mechanism: Massey WF2 model. Peak doping $\sim 2 \times 10^{16} \text{ cm}^{-3}$.
3. Bias above gain depletion voltage (~ 30 –50 V extra).
4. **Set** $\rightarrow$ **Potentials** $\rightarrow$ verify gain layer in the 1D cut (pad 4 of Drift Potential tab).
5. **Currents** $\rightarrow$ observe enhanced charge (gain electrons, gain holes) in the Charge Collection panel.

12.3 Irradiation Study

1. Configure detector as above.
2. Enable **All On/Off** in the Irradiation frame.
3. Set Fluence = 100 [$10^{14} \text{ n}_{\text{eq}}/\text{cm}^2$].
4. Compare collected charge vs. unirradiated case.
5. Increase gain layer doping to compensate for gain quenching if desired.

12.4 Batch Run for Time Resolution

1. Select particle: MIP Landau.
2. Enable **Batch Mode**, set e.g. 1000 events.
3. Enable **Electronics ON**, configure amplifier (e.g. FAST3 preset).
4. Press **Currents**.
5. Read time resolution from the jitter histogram in the Electronics I tab.

13 Physical Constants and Default Parameters

Constant	Symbol	Value	Units
Vacuum permittivity	ε_0	8.854×10^{-12}	F/m
Si relative permittivity	ε_r	11.9	—
Elementary charge	e	1.602177×10^{-19}	C
Boltzmann constant	k_B	1.38065×10^{-23}	J/K
Electron mass	m_0	9.109383×10^{-31}	kg
Grid unit	—	1	μm
Si ionization energy	E_{ion}	3.6	eV/pair
SiC ionization energy	E_{ion}	8.0	eV/pair
Alpha range in Si (5 MeV)	—	~ 30	μm
Laser 1064 nm (Si)	—	75	e-h/ μm
Si electron mobility (300 K)	μ_e	0.1417	$\text{m}^2/(\text{V}\cdot\text{s})$
Si hole mobility (300 K)	μ_h	0.0470	$\text{m}^2/(\text{V}\cdot\text{s})$
Si electron saturation velocity	$v_{\text{sat},e}$	1.07×10^5	m/s
Si hole saturation velocity	$v_{\text{sat},h}$	8.37×10^4	m/s
Diamond electron mobility	μ_e	0.18	$\text{m}^2/(\text{V}\cdot\text{s})$
Diamond hole mobility	μ_h	0.15	$\text{m}^2/(\text{V}\cdot\text{s})$
Diamond saturation velocity	v_{sat}	2.7×10^5	m/s
SiC electron mobility	μ_e	0.08	$\text{m}^2/(\text{V}\cdot\text{s})$
SiC hole mobility	μ_h	0.0115	$\text{m}^2/(\text{V}\cdot\text{s})$
SiC saturation velocity	v_{sat}	2.7×10^5	m/s

Massey impact ionization model (silicon):

$$\alpha_e = 4.43 \times 10^5 \exp\left(-\frac{9.66 \times 10^5}{E}\right) \quad [\text{cm}^{-1}] \quad (25)$$

$$\alpha_h = 1.13 \times 10^5 \exp\left(-\frac{1.97 \times 10^6}{E}\right) \quad [\text{cm}^{-1}] \quad (26)$$

where E is in V/cm.

Built-in technology gain layer profiles:

Technology	Depth start	Depth end
FBK standard	0.5 μm	1.0 μm
FBK LD	0.6 μm	1.0 μm
FBK LD Deep	1.4 μm	1.8 μm
CNM	0.0 μm	2.5 μm
HPK standard	1.3 μm	1.9 μm
HPK LD	1.1 μm	1.5 μm
HPK LD Deep	1.9 μm	2.3 μm

This manual was written for Weightfield2 Build 8.0 “The Claude Edition”.
Based on direct analysis of the source code in February 2026.